

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Hair■care products exhibit the highest average defect rate (2.48 %) while Carrier C delivers the strongest revenue■to■shipping■cost performance (\$1,138.56 per shipping dollar). Thirty■six SKUs failed inspection, with several skin■care items (e.g., SKU42, SKU65) exceeding a 4.9 % defect rate. Stock■out exposure is acute for a handful of items, notably SKU34 (skincare) whose risk score of 13 reflects a single unit on hand against a 26■day lead time.

****KEY FINDINGS**** - ****Product■type defect hierarchy:**** Hair■care = 2.483 % (defect), Skin■care = 2.335 %, Cosmetics = 1.919 % (lowest). - ****Carrier efficiency:**** Carrier C generates the highest revenue per shipping dollar (\$1,138.56), followed by Carrier B (\$1,055.70) and Carrier A (\$917). Carrier C also maintains a modest avg. defect rate (2.6 %) relative to its revenue impact. - ****Failed■inspection SKUs:**** 36 SKUs flagged; the top three defect rates are SKU42 (4.94 %, skin■care, stock 22), SKU65 (4.91 %, skin■care, stock 57) and SKU50 (4.75 %, cosmetics, stock 31). All failed SKUs span all three product categories and three carriers. - ****Stock■out risk concentration:**** SKU34 (skincare) scores 13 – the highest risk (stock 1, lead■time 26 days). SKU68 (hair■care) scores 8 with zero inventory and an 8■day lead■time. The next highest scores (>3) are SKU24, SKU2, SKU33, SKU47, SKU57. - ****Revenue contribution:**** Skin■care drives the largest revenue (\$241,628), followed by Hair■care (\$174,455) and Cosmetics (\$161,521). Gross■margin ratios exceed 99 % across all categories, indicating strong pricing power but masking quality■related cost exposures.

****RISKS AND RECOMMENDATIONS**** The elevated defect rate in hair■care and the concentration of high■defect SKUs (especially in skin■care) pose a quality■risk to brand reputation; prioritize root■cause analysis with the affected carriers (Carrier C and Carrier A) and enforce tighter incoming inspection thresholds. Simultaneously, mitigate imminent stock■outs by increasing safety stock for the top■risk SKUs (e.g., SKU34, SKU68) and reviewing lead■time reductions or alternative sourcing to align inventory with demand.